

Source Material Review

Annex to the test assignment response · Plata · Group IRM · 12 August 2026
What was found in Appendices A–I, the assignment text and the interview notes

Why this document exists. The assignment hands over the appendices as "day-one documentation" and asks for an assessment of the landscape. You cannot assess the landscape without reading the documentation itself: if the day-one document contains contradictions, then those contradictions — not the data — are the first thing a new analyst will hit. What follows is a full read-through: **20 findings**, each with a quote and a file and line reference. Some were found by reading, some by running the documented queries against reproduced data.

20

findings in the
source materials

8

rated critical

5

links to files
that do not exist

3

found only by
running queries

Summary by class

Class	Findings	Why it is dangerous
A. Contradictions between documents	6	The reader picks one of two versions and does not know which one they picked
B. Omissions and dangling links	5	The schema map is incomplete; half the cross-references lead nowhere
C. Examples that do not work as described	4	A query copied from the documentation returns the wrong result
D. Subject-matter errors	3	The calendar and the reference data are factually wrong
E. Methodology	2	The statistics understate or overstate significance

A. Contradictions between documents

#	Where	What it says	The problem	Severity
A1	Appendix B, lines 46 / 74 / 79	"Data freshness: T-1 (calendar days)" — and in the table below: "portfolio_cor ... today - 2 " — and further down: "portfolio_cor is typically T-1"	Three statements about one table, two of them incompatible. The SLA threshold depends on the choice: monitoring at T-1 will alert every day, monitoring at T-2 will miss a full day of delay	critical
A2	Appendix B vs F	B: statement_id in portfolio_cor is TEXT . F: statement_id is Integer , sourced from ods.pf_engine.STATEMENTS	This is the join key between layers. Different types produce either a cast error or an implicit cast that loses matches	critical
A3	Appendix C, line 106 vs D, line 62	C: $\text{SUM}(\text{cor_prnp}) / \text{SUM}(\text{avg_balance_prnp_net}) * 12$, labelled "Correct COR% by vintage". D: $-\text{SUM}(\text{a.cor_prnp}) / \dots$	C has no minus sign, while cor_prnp is negative by the convention stated in Appendix C itself. An example labelled as correct returns a negative COR%. The same query also lacks the product filter that D has	critical
A4	Appendix E, line 22 vs G, line 55	E: the EL LT formula is $\text{cor_predict_lt} / \text{bal_predict_lt}$. G: $\text{SUM}(\text{cor_total_predict_lt}) / \text{SUM}(\text{bal_total_predict_lt})$	The columns are named differently. The glossary version will not compile	substantial
A5	Assignment text vs Appendix A	Assignment: "20+ schemas across 6 databases ". Appendix A lists nine distinct database prefixes	A discrepancy in the framing. The map contains 19 schemas — "20+" only holds if the unlisted ods.ref is added (see B1)	minor
A6	Assignment, Part 2 vs Appendix H	In the three-calculation table both Risk Analytics and Portfolio Monitoring are labelled "Gen3", differing only in formula and annualization	Per Appendix H, the cor column in the challenger table excludes the 0-DPD bucket , whereas cor_prnp in the statement table does not. These are different numerators, not one model. Measured, the divergence contributes 0.2–0.5 pp	substantial

B. Omissions and dangling links

#	Where	What	The problem	Severity
B1	Appendix A	The <code>ods.ref</code> schema is absent from the schema map (0 mentions), yet its <code>HOLIDAYS</code> table is documented in detail in Appendix B (3 mentions)	The schema map that both humans and the agent navigate by is incomplete. And what is missing is not some technical schema but the calendar on which every business-day calculation depends	critical
B2	Appendix B	<code>ods.ref.HOLIDAYS</code> is documented inside the file "DM Schema — Table Inventory"	An ODS table described in a mart inventory. Anyone looking for it by layer will not find it	minor
B3	A, D, G, H	References to <code>tables_technical.md</code> , <code>tables_dm.md</code> , <code>finance/gl_tables.md</code> , <code>finance/mxgaap_provisions.md</code> , <code>risk/pilots.md</code>	Five links, not one file. The appendices are extracts from a larger knowledge base, and the cross-references broke during extraction. For an agent such a link is a dead end with no error signal	substantial
B4	Appendix H, line 32	The formula <code>cor_total = cor_dlq_fact + cor_0_bucket</code> , while <code>cor_dlq_fact</code> and <code>cor_0_bucket</code> are absent from the key-column list of that same table	The formula references columns that, per the documentation, do not exist. Additivity cannot be verified from the document	substantial
B5	Appendix H	"This decomposition avoids the known cor_collection_fact data issue "	A known data defect is mentioned in a single phrase and described nowhere: not what it consists of, not how large it is, not when it will be fixed. The workaround is documented, the cause is not	critical

C. Examples that do not work as described

These findings came not from reading but from **running the documented queries against reproduced data**. They are invisible to the eye.

#	Where	What is claimed	What actually happens	Severity
C1	Appendices A and D	"Using the wrong filter column returns no results or wrong data silently "	In Snowflake, referencing a non-existent column raises a compilation error , not an empty result. The miss can only be silent if the column physically exists — for example, if it survived a rename. The symptom is described correctly, the mechanism is not, and the description does not tell you what to look for	substantial
C2	Appendix C, line 106	The query is labelled " - - Correct COR% by vintage "	It returns a negative percentage: the minus in front of the sum is missing. An analyst who copies the example gets a COR with a negative sign and will either flip it by hand or build the chart upside down	critical
C3	Appendix B, the "business days in a month" query	Returns <code>month_start</code> as the month	It returns <code>TIMESTAMP_NTZ</code> rather than <code>DATE</code> : the string literal <code>'2026-01-01'</code> inside <code>DATEADD</code> is cast to a timestamp. Verified — joining the result on a <code>DATE</code> column yields no matches. The query itself runs; the error surfaces further down the chain	substantial
C4	Appendix B, line 77	The freshness table lists <code>dm_npv_gen5p_details</code>	No such object exists: everywhere else it is <code>dm.npv_gen5_details</code> . The stray <code>p</code> plus an underscore instead of a dot is probably a remnant of the former name <code>NPV_GEN5_PV_DETAILS</code> , which per Appendix G has been dropped	minor

D. Subject-matter errors

#	Where	What it says	What is wrong	Severity
D1	Appendix B, line 97	Among the holidays: "Inauguración (Dec 1 , every 6 years)"	The transfer of executive power in Mexico moved from 1 December to 1 October , effective 2024. A calendar built from this description will place a non-business day in the wrong month once every six years and will miss the quarter in which write-offs fall	substantial
D2	Appendix B, line 97	In the same list: "Día de las Madres (May 10)"	10 May is not a mandatory non-working day under Article 74 of the Federal Labour Law and, as far as I am aware, is not part of the CNBV banking calendar. An extra non-business day understates the number of business days in May and distorts the annualization of COR. This needs reconciliation against the official CNBV calendar — as does the whole list: it mixes statutory public holidays with banking ones (Semana Santa, 2 November, 12 December) without stating which source is authoritative	critical
D3	Appendix E, line 9	"BdC Buro de Credito Mexican credit bureau (Circulo de Credito)"	Buró de Crédito and Círculo de Crédito are two different Mexican credit bureaus, not one entity with two names. The glossary line conflates them. Further down the same file, "BDC Trans Union Califica" adds an inconsistent spelling of the abbreviation	substantial

E. Methodology

#	Where	What it says	The problem	Severity
E1	Appendix D, t-statistic	<code>cor%_std = cor_forecast% × sqrt(var_cor/cor² + var_bal/bal²)</code> , labelled "delta method for ratio variance estimation"	The delta method for a ratio has a third term — the covariance of numerator and denominator. It is omitted here, i.e. COR and balance are treated as independent. They are strongly positively related, so the variance is overstated and the t-statistic understated: deviations that are significant will look insignificant. The independence assumption is not stated in the text	substantial
E2	Appendix D	The SQL example computes <code>actual_cor_pct</code> and <code>forecast_cor_pct</code> as positive . The Tableau formulas alongside make both negative . The t formula uses the Tableau variant	Two sign conventions inside one appendix. Substituting the numbers from the SQL example into the t formula flips the sign — producing the conclusion "worse than plan" instead of "better than plan"	critical

F. Observations that are not errors

- **Priority is not set by effect size.** The rule "filter on `statement_num >= 2`" is repeated verbatim **three times** — in Appendices C, D and F. Measured on reproduced data, its contribution to the divergence between calculations is 0.05–0.24 pp, whereas the contribution of the grain choice is 2.1–2.7 pp — an order of magnitude larger, and mentioned nowhere. The warning is right, its weighting is not.
- **The interview notes contain speech-recognition artefacts** and are not marked as a draft: "Axon EDC Disk Group", "diplomint coordinator", "Wisla/Voyager", "anpoints", "IChart, ISIT". Several employment periods are recorded as

"unknown". To a human this is noise; to an LLM it is a source from which it will reproduce non-existent product names as fact. Such files should either be proofread or explicitly flagged as "machine transcript, not a source of truth".

- **Appendix I uses DMBOK loosely.** It says "across the 11 DMBOK disciplines" and names Change Management and Data Catalog Management among the six tracked. DAMA DMBOK has eleven knowledge areas and neither of those is among them: the catalog falls under Metadata Management, change control under Data Governance. Reference and Master Data are **one** area in DMBOK and are split into two here. The assignment itself asks to "connect your proposals to DMBOK disciplines", so the terminology should either be aligned to the framework or honestly labelled as an in-house adaptation.

How each class is caught systematically

The main observation from the read-through: **the findings fall into classes, and each class needs its own detector**. No single tool closes all twenty, and trying to close them all with an "attention to detail" review does not scale and rests on one person.

Class of finding	Detector	Mechanics	What it closes from the list
Document vs reality columns and tables that do not exist	dbt + linter , deterministic	Every name in the YAML descriptions is checked against <code>INFORMATION_SCHEMA</code> ; the test fails on a mismatch	A2, A4, B1, B4, C4
SQL examples in the documentation	CI , deterministic	Every code block in the documentation is executed against reference data and its result compared with the expectation. Examples become tests	C2, C3 — and this is exactly how three of my findings were made
Document vs document contradictions between files	LLM validator	Poorly caught by code: it requires understanding that "T-1" and "today – 2" describe the same thing. The agent is given an explicit task: "find statements about the same object that are incompatible"	A1, A3, A5, A6, E2
Schema drift	DDL snapshot diff	The DDL hash is compared with the snapshot and with the change log. A changed hash with no log entry is an unregistered change	the C1 class: a column that survived a rename
Data drift same schema, different numbers	data diff	Row-level comparison of model versions (<code>audit_helper.compare_relations</code>). Catches a model re-estimation — the class where nothing breaks	the whole Part 3 scenario
Subject-matter facts calendar, reference data	reconciliation with an external source	The calendar is not hand-written but loaded from the authoritative source (CNBV) and reconciled annually	D1, D2
Methodology	a human , with the LLM prompting	The agent flags "assumption not stated"; the expert makes the call	E1

What dbt specifically provides

A significant share of what I assembled by hand in the sandbox is replaced in production by native dbt mechanisms — and that is more honest than maintaining my own:

Built by hand	dbt equivalent
26 assertions in the <code>BUILD_ASSERTIONS</code> view	generic tests (<code>unique</code> , <code>not_null</code> , <code>accepted_values</code> , <code>relationships</code>) plus singular tests for semantics: sign in the aggregate, additivity, absence of weekends, zero COR on the first statement
Consumer registry	<code>exposures</code> — with type, owner and dependencies; change impact is computed from the graph rather than from a table
Table and column comments	<code>persist_docs</code> — descriptions live in YAML next to the model and are pushed to Snowflake automatically . One source instead of two
DDL snapshot and drift detector	<code>manifest.json</code> artefacts + <code>state:modified</code> : CI shows exactly what changed before it is applied
Object catalog	<code>dbt docs</code> — with a caveat: it covers models, not the whole warehouse. The ODS layer still has to be documented separately

What dbt does **not** close: contradictions between prose documents, subject-matter errors in reference data, and anything outside the model graph. That remainder is exactly what the agent is for.

What the validator agent should look like

The key requirement: **the validator is a separate role from the analyst**. They have different jobs, different privileges and a different cost of error. The analyst answers questions and should be cautious; the validator hunts for discrepancies and

should be pedantic.

1. **Inputs:** the documentation, `INFORMATION_SCHEMA`, the catalog, the change log and test results. Not text alone — otherwise there is nothing to reconcile against.
2. **Four explicit tasks** instead of a generic "check this": (1) execute every code example; (2) reconcile every object and column mentioned against reality; (3) find statements about the same object that contradict each other; (4) flag unstated assumptions.
3. **Output:** findings with file, line and quote — as in this document. Without a quote a finding is unverifiable and will be ignored.
4. **Human triage of false positives.** A validator whose findings are not filtered teaches the team to ignore it within two weeks — exactly like the Monday false freshness alarm.
5. **The validator has its own golden set.** Otherwise there is no one to validate the validator: feed it a document with twenty known defects and see how many it finds. This document is a ready-made set — twenty labelled findings with answers.

What I would put into CI in the first week. Ordered by benefit-to-effort, not by completeness.

1. Execution of every code example from the documentation against reference data. Cheapest of all, and it catches the most dangerous class — "copied from the document and got the wrong number".
2. Reconciliation of documented names against `INFORMATION_SCHEMA`. Catches A2, A4, B1, B4 and C4 with a single test.
3. DDL snapshot and reconciliation against the change log — already built and working.
4. LLM review for contradictions — weekly on a schedule rather than on every commit: it is the most expensive and the noisiest detector.

What I propose to do with the appendices themselves

Do not edit them. The appendices were issued as the terms of the assignment, and editing the terms to fit one's own implementation is a substitution. The right order is:

1. **The eight critical findings** (A1, A2, A3, B1, B5, C2, D2, E2) go into tickets raised against the documentation owner, with quotes — that is first-week work, not a matter for debate.
2. **The negative COR% in Appendix C and the sign confusion in the t-statistic** should be fixed first: they produce a wrong number silently.
3. **The holiday calendar** should be reconciled with the official CNBV source and moved from a hand-written list to a load.
4. **The interview notes** should be marked as a machine transcript — one line in the file header, against a high cost of error if an agent reads them.

Every quote is verifiable: file and line number are given for each finding. Three class C findings were obtained by running the documented queries against the sandbox `snowflake-sandbox/`; the rest by reading.